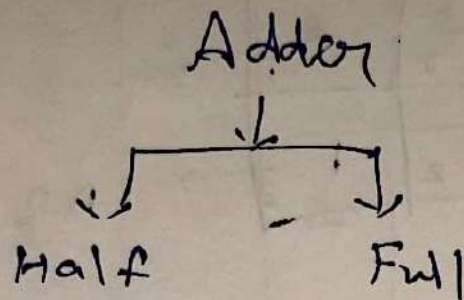
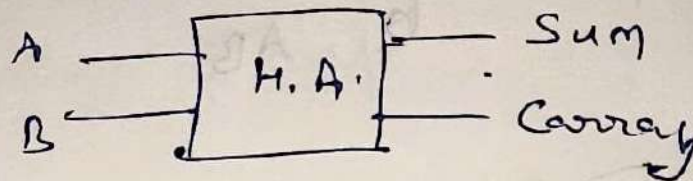


## Unit - 4

### \* Combinational circuit \*

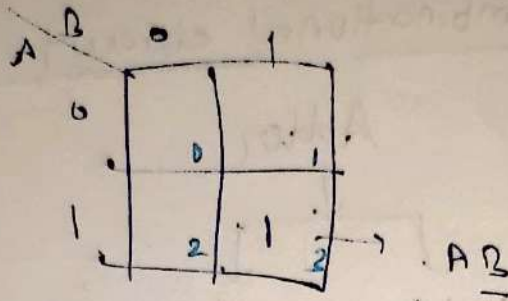


### Half Adder

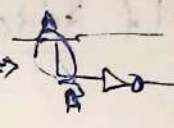


| I/P |   | O/P |       |
|-----|---|-----|-------|
| A   | B | Sum | Carry |
| 0   | 0 | 0   | 0     |
| 0   | 1 | 1   | 0     |
| 1   | 0 | 1   | 0     |
| 1   | 1 | 0   | 1     |

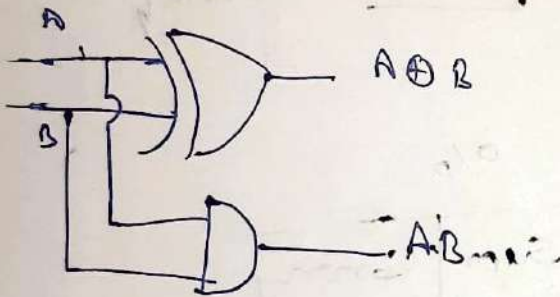
For Carry



eq.  $\Rightarrow AB$

Now Both eq.  $A \oplus B \Rightarrow$  

$AB$



Full Adder



| I/p |   |     | Output |      |
|-----|---|-----|--------|------|
| A   | B | Cin | Sum    | Cout |
| 0   | 0 | 0   | 0      | 0    |
| 0   | 0 | 1   | 1      | 0    |
| 0   | 1 | 0   | 1      | 0    |
| 0   | 1 | 1   | 0      | 1    |
| 1   | 0 | 0   | 1      | 0    |
| 1   | 0 | 1   | 0      | 1    |
| 1   | 1 | 0   | 0      | 1    |
| 1   | 1 | 1   | 1      | 1    |

For Sum Kmap

| BC <sub>in</sub> | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| A                |    |    |    |    |
| 0                |    | 1  | 1  | 1  |
| 1                | 1  |    | 1  |    |

$$A \oplus B \oplus C_{in}$$

For Carry Kmap

| BC <sub>in</sub> | 00 | 01 | 11 | 10 |
|------------------|----|----|----|----|
| A                |    |    |    |    |
| 0                |    |    | 1  | 1  |
| 1                |    | 1  | 1  | 1  |

BC

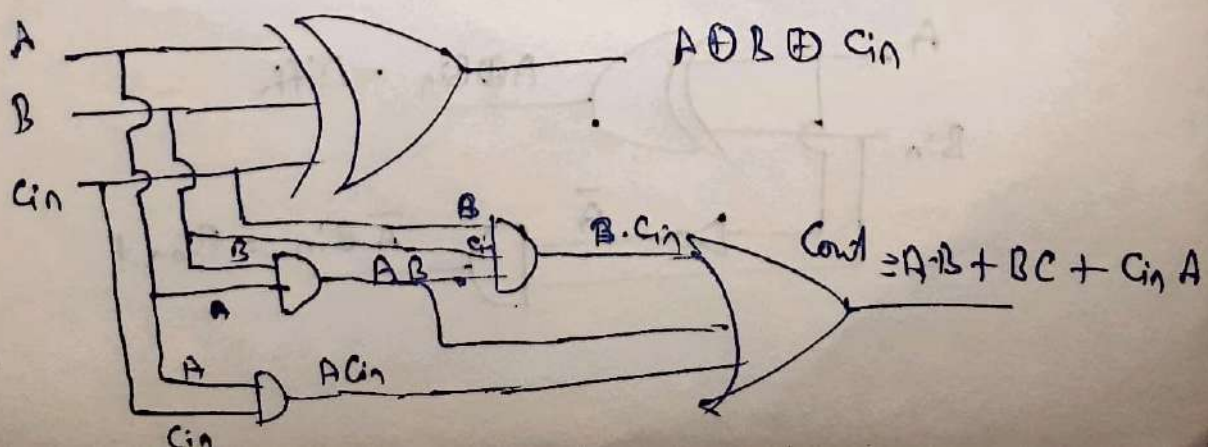
A B

A C<sub>in</sub>

$$C_{out} = AB + BC_{in} + AC_{in}$$

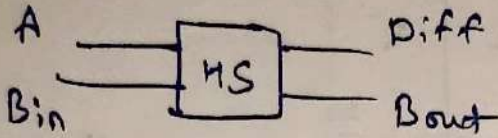
Now Both eq.

Circuit.



# Subtraction Circuit

Half



$B \Rightarrow$  Borrow

| A | $B_{in}$ | Diff | Bout |
|---|----------|------|------|
| 0 | 0        | 0    | 0    |
| 0 | 1        | 1    | 1    |
| 1 | 0        | 1    | 0    |
| 1 | 1        | 0    | 0    |

for Diff K map

| A \ $B_{in}$ | 0 | 1 |
|--------------|---|---|
| 0            | 0 | 1 |
| 1            | 1 | 0 |

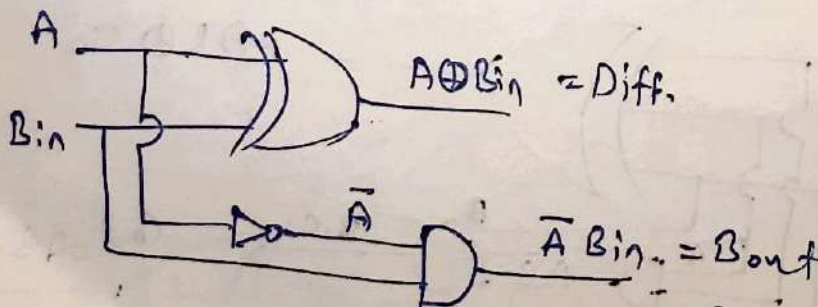
$A \oplus B_{in}$

for Borrow K map

| A \ $B_{in}$ | 0 | 1 |
|--------------|---|---|
| 0            | 0 | 1 |
| 1            | 2 | 2 |

$\Rightarrow \bar{A} B_{in}$

Now Circuit



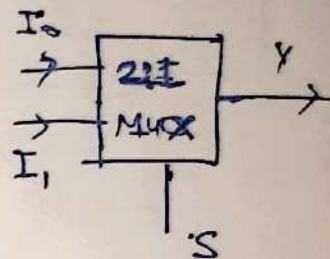
Multiplexer :- Multiplex means many into one. A Digital multiplexer

(MUX) is a combinational circuit have many input ~~digit~~ line to single output line. Output is depend on selection line.

Input =  $2^m$  where  $m$  = selection line

(i) 2 to 1 multiplexer -

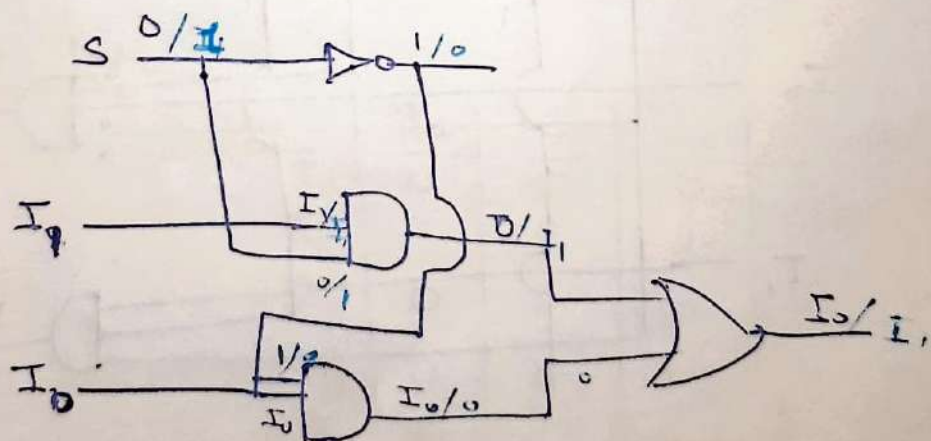
Input = 2 output = 1  
then  $m = 1$



(ii) Truth table

| S | Y     |
|---|-------|
| 0 | $I_0$ |
| 1 | $I_1$ |

\* Circuit Diagram



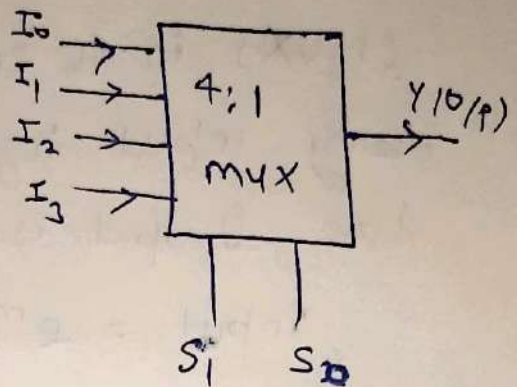
(ii) 4:1 multiplexer

input = 4

$$2^m = 4$$

$$m = 2$$

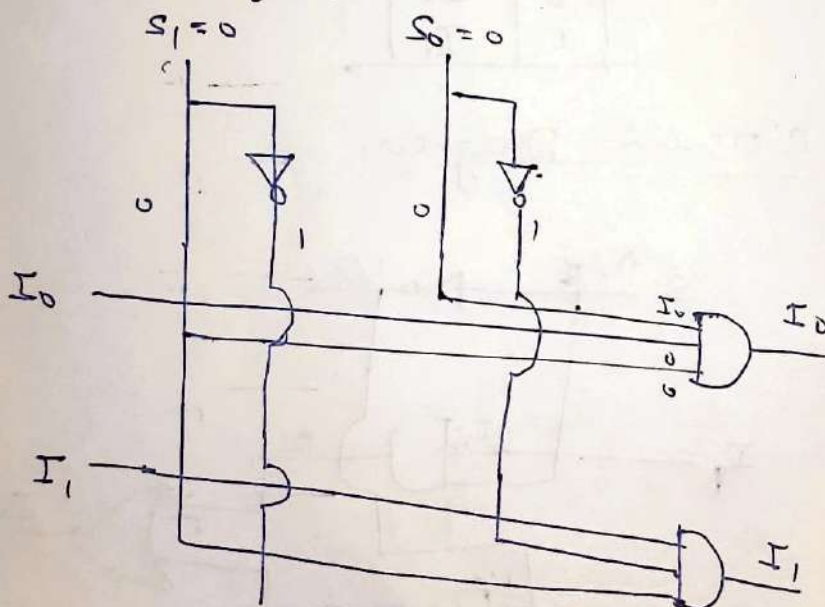
Selection line = 2

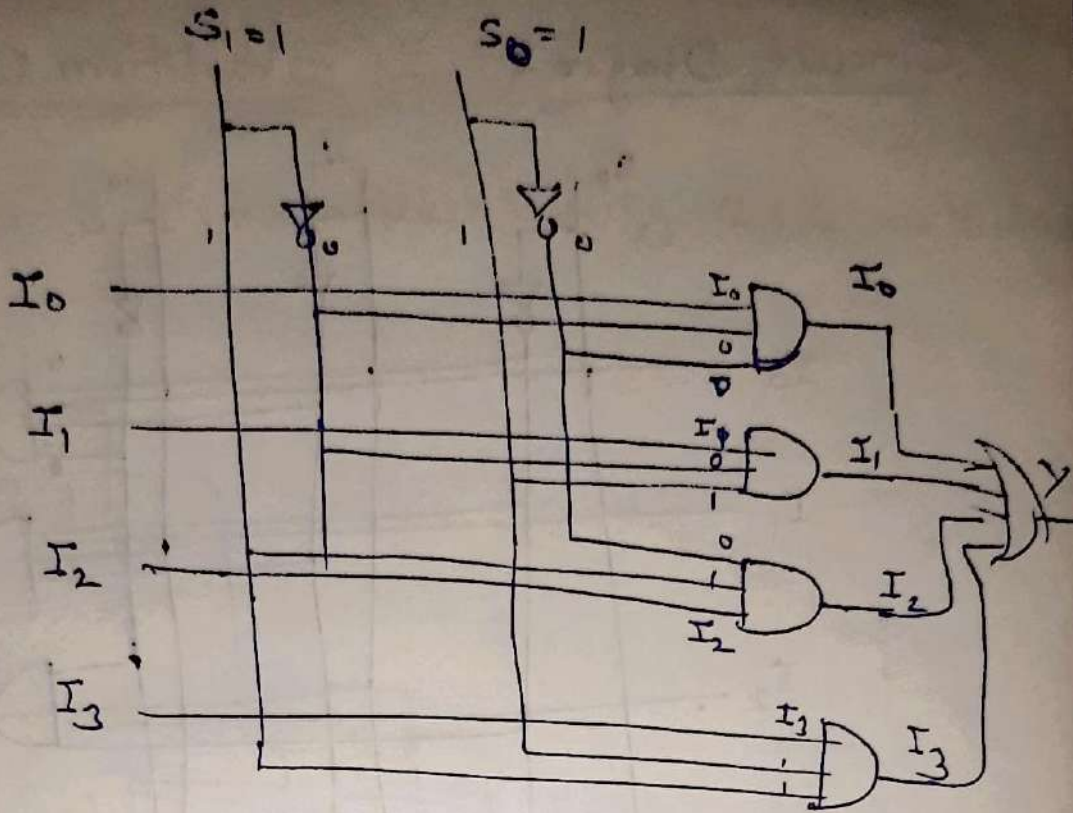


Truth table

| $S_1$ | $S_0$ | $Y$   |
|-------|-------|-------|
| 0     | 0     | $I_0$ |
| 0     | 1     | $I_1$ |
| 1     | 0     | $I_2$ |
| 1     | 1     | $I_3$ |

Circuit Diagram

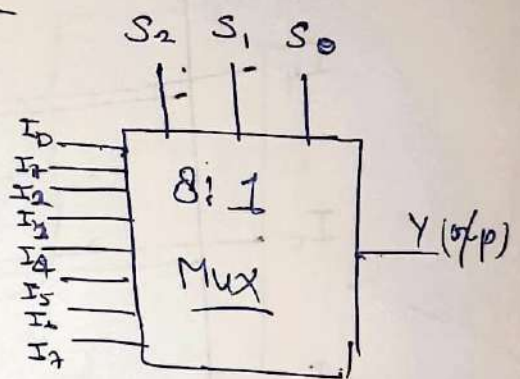




i) 8:1 Multiplexer :-

$2^m = 8$  input

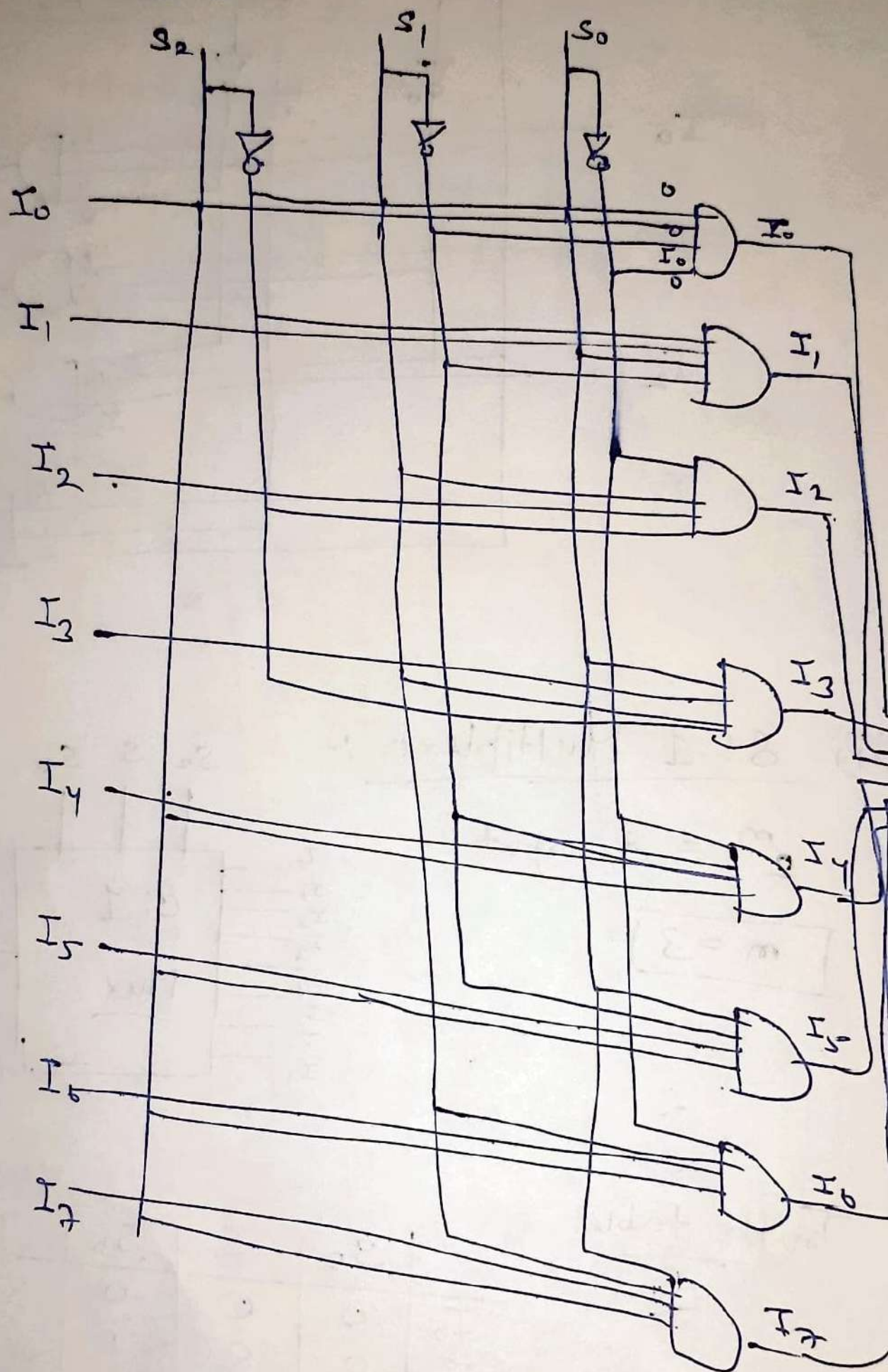
$$m = 3$$



Truth table

|       | $S_2$ | $S_1$ | $S_0$ |
|-------|-------|-------|-------|
| $I_0$ | 0     | 0     | 0     |
| $I_1$ | 0     | 0     | 1     |
| $I_2$ | 0     | 1     | 0     |
| $I_3$ | 0     | 1     | 1     |
| $I_4$ | 1     | 0     | 0     |
| $I_5$ | 1     | 0     | 1     |
| $I_6$ | 1     | 1     | 0     |
| $I_7$ | 1     | 1     | 1     |

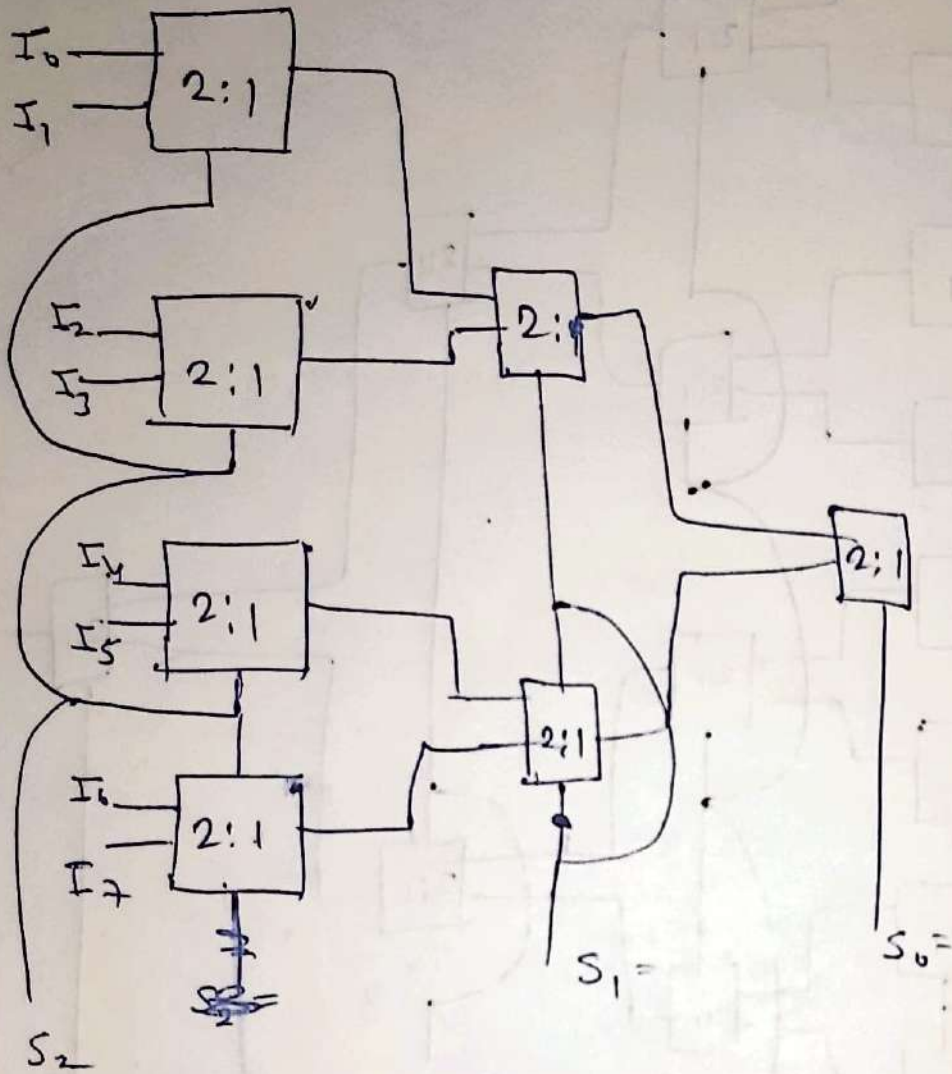
# Circuit Diagram



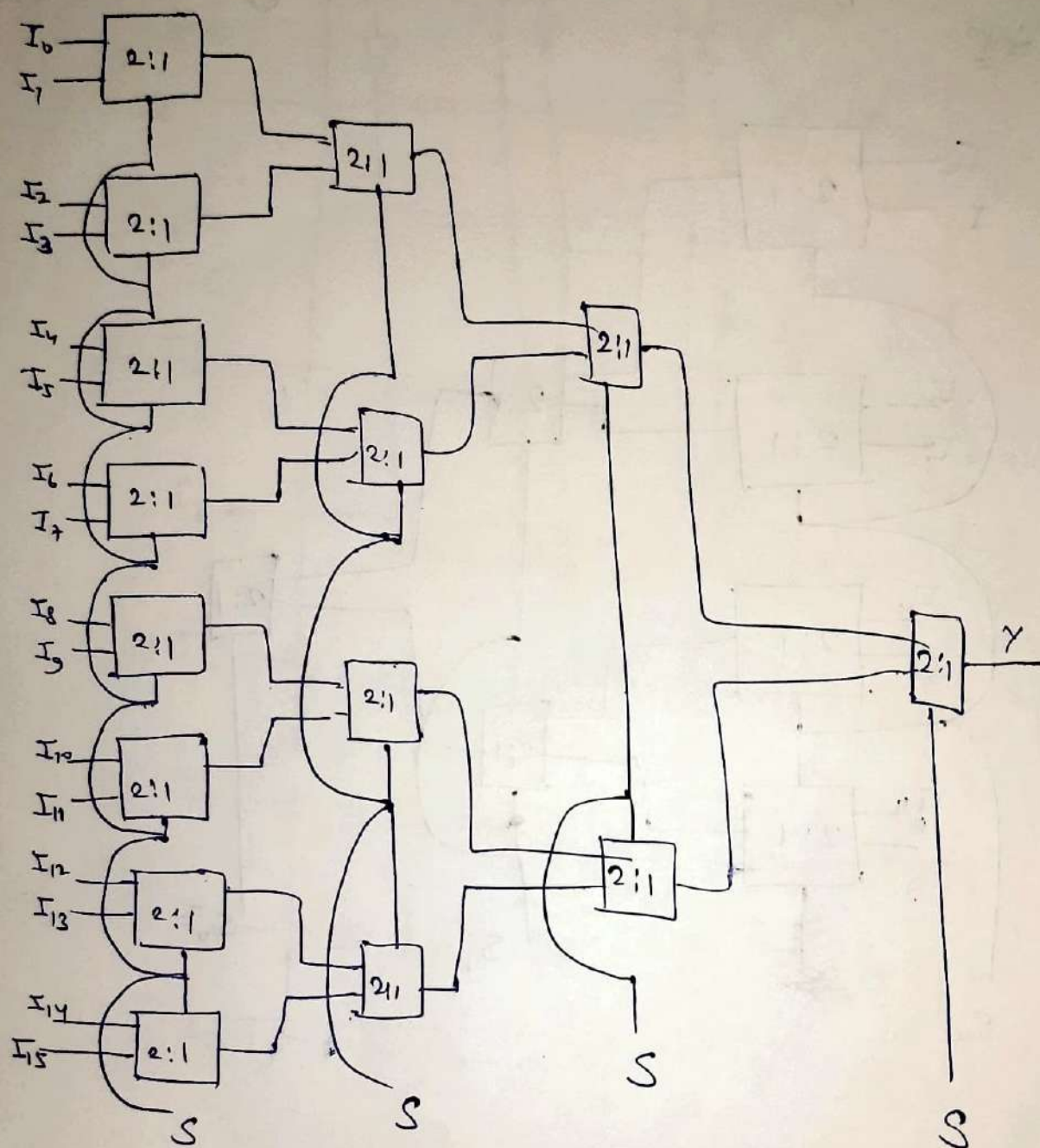
## Type - 0 multiplexer :-

Q. Design 8:1 multiplexer using 2:1 multiplexer

Sol<sup>n</sup>



Q Design 16:1 multiplexer using 2:1



Q Design 16:1 multiplexer using 4:1 multiplexer.

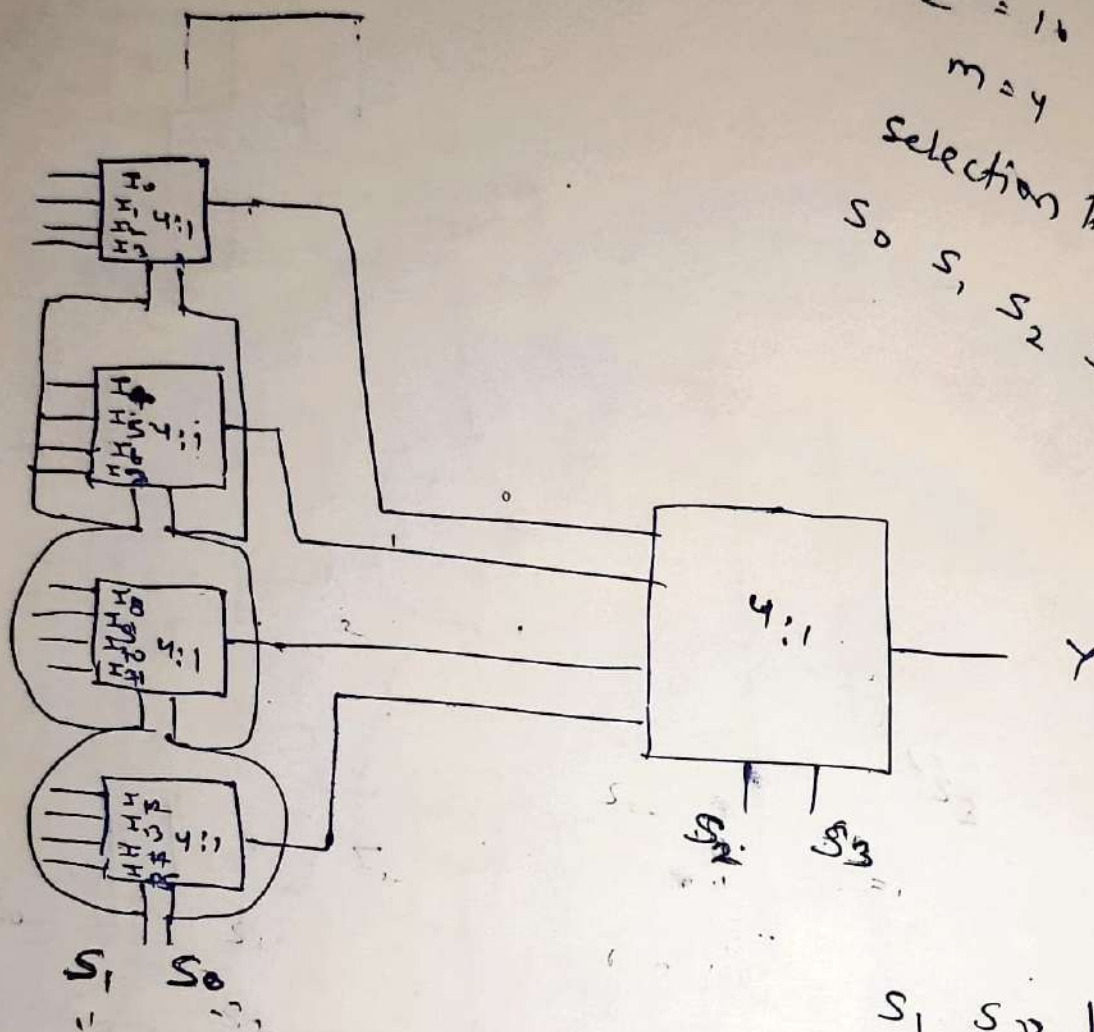
Soln

$$2^4 = 16$$

$$m = 4$$

Selection line = 4

$S_0, S_1, S_2, S_3$



| $S_1$ | $S_0$ | $Y$ |
|-------|-------|-----|
| 0     | 0     | $H$ |
| 0     | 1     | $H$ |
| 1     | 0     | $H$ |
| 1     | 1     | $H$ |

## \* Type - 1 multiplexer :-

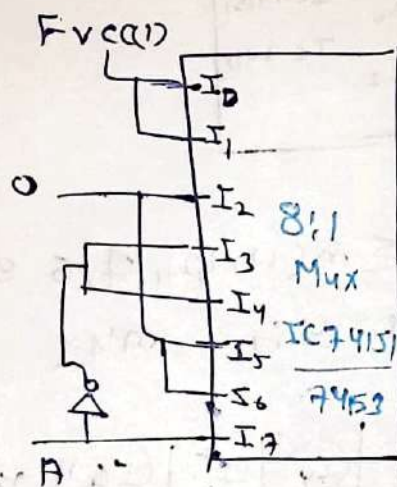
### ● Implementation of boolean expression using multiplexer

Q  $F(A, B, C, D) = \sum (0, 1, 3, 4, 8, 9, 15)$

number of variable = selection line: = 4

Input =  $2^4 = 16$

|           |       |       |          |           |           |          |          |          |
|-----------|-------|-------|----------|-----------|-----------|----------|----------|----------|
| $\bar{A}$ | $I_0$ | $I_1$ | $I_2$    | $I_3$     | $I_4$     | $I_5$    | $I_6$    | $I_7$    |
| A         | $I_8$ | $I_9$ | $I_{10}$ | $I_{11}$  | $I_{12}$  | $I_{13}$ | $I_{14}$ | $I_{15}$ |
|           | 1     | 1     | 0        | $\bar{A}$ | $\bar{A}$ | 0        | 0        | A        |

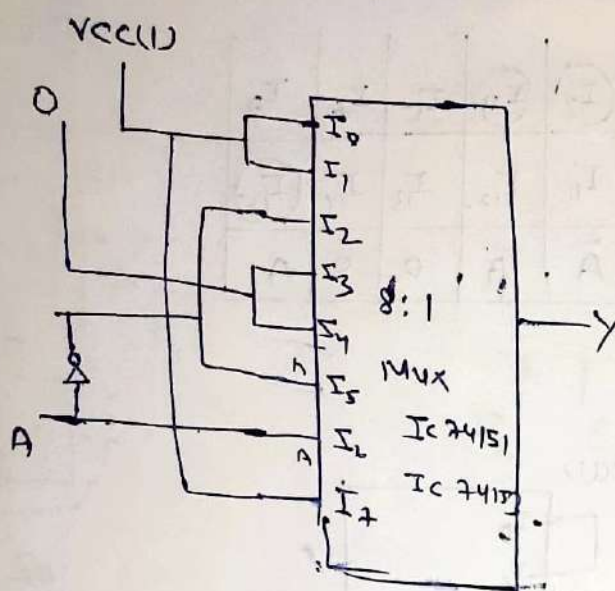


Q

$$F(A, B, C, D) = \sum m(0, 1, 2, 5, 7, 8, 9, 14, 15)$$

using 8:1 mux

| $\bar{A}$ | $I_0$ | $I_1$ | $I_2$     | $I_3$    | $I_4$    | $I_5$     | $I_6$    | $I_7$    |
|-----------|-------|-------|-----------|----------|----------|-----------|----------|----------|
| A         | $I_8$ | $I_9$ | $I_{10}$  | $I_{11}$ | $I_{12}$ | $I_{13}$  | $I_{14}$ | $I_{15}$ |
|           | 1     | 1     | $\bar{A}$ | 0        | 0        | $\bar{A}$ | A        | 1        |



Q

$$F(A, B, C, D) = \sum m(0, 1, 2, 4, 6, 9, 12, 14)$$

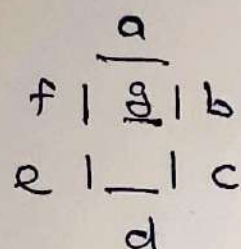
using 4:1 mux

| $\bar{A}$ | $I_0$     | $I_1$ | $I_2$     | $I_3$    | $I_4$    | $I_5$    | $I_6$    | $I_7$    |
|-----------|-----------|-------|-----------|----------|----------|----------|----------|----------|
| A         | $I_8$     | $I_9$ | $I_{10}$  | $I_{11}$ | $I_{12}$ | $I_{13}$ | $I_{14}$ | $I_{15}$ |
|           | $\bar{A}$ | 1     | $\bar{A}$ | 0        | 1        | 0        | 1        | 0        |

# BCD to seven Segment Decoder :-

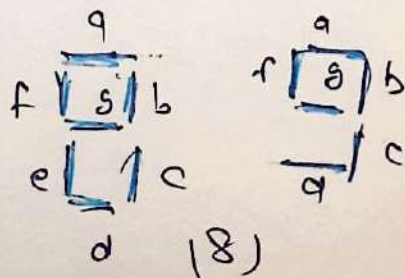
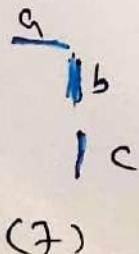
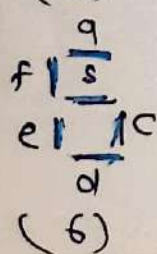
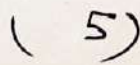
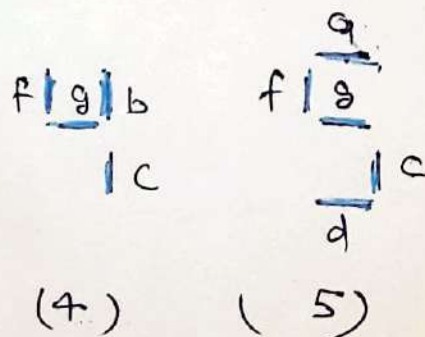
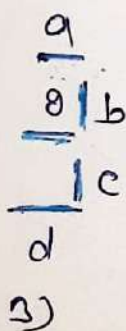
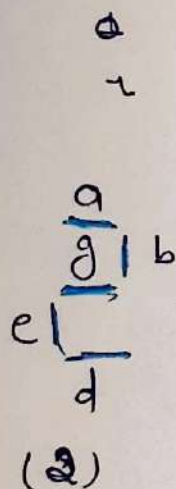
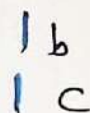
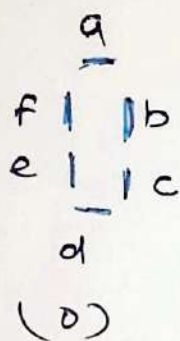
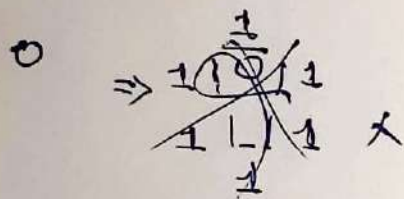
A seven segment display is normally used for displaying one of the decimal digit 0 through 9.

A seven segment display composed of seven element of segment. Each segment is made up of a material that emit light when current is passed through it.



if  $b \& c \Rightarrow 1$  to all zero  
then high  $\Rightarrow 9$

if  $g$  is zero & all 1  
then low  $\Rightarrow 0$

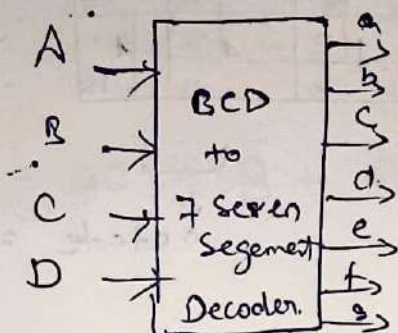


\* Design BCD to 7 segment decoder.

Number of output  $\Rightarrow a, b, c, d, e, f, g$

input  $\Rightarrow 2^4 \Rightarrow (9) \checkmark$

(input < output)



| Input |   |   |   |   | Output |   |   |   |   |   |   |
|-------|---|---|---|---|--------|---|---|---|---|---|---|
|       | A | B | C | D | a      | b | c | d | e | f | g |
| 0     | 0 | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 0 |
| 1     | 0 | 0 | 0 | 1 | 0      | 1 | 1 | 0 | 0 | 0 | 0 |
| 2     | 0 | 0 | 1 | 0 | 1      | 1 | 0 | 1 | 1 | 0 | 1 |
| 3     | 0 | 0 | 1 | 1 | 1      | 1 | 1 | 1 | 0 | 0 | 1 |
| 4     | 0 | 1 | 0 | 0 | 0      | 1 | 1 | 0 | 0 | 1 | 1 |
| 5     | 0 | 1 | 0 | 1 | 1      | 0 | 1 | 1 | 0 | 1 | 1 |
| 6     | 0 | 1 | 1 | 0 | 1      | 0 | 1 | 1 | 1 | 1 | 1 |
| 7     | 0 | 1 | 1 | 1 | 1      | 0 | 1 | 0 | 0 | 0 | 0 |
| 8     | 1 | 0 | 0 | 0 | 1      | 1 | 1 | 1 | 1 | 1 | 1 |
| 9     | 1 | 0 | 0 | 1 | 1      | 1 | 1 | 1 | 0 | 1 | 1 |

$$g = \Sigma(0, 2, 3, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$a = \Sigma(0, 2, 3, 4, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$b = \Sigma(0, 1, 2, 3, 4, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$c = \Sigma(0, 1, 3, 4, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$$

$$d = \Sigma(0, 2, 3, 5, 6, 8, 9) + d(11, 12, 13, 14, 15)$$

$$e = \Sigma(0, 2, 6, 8) + d(11, 12, 13, 14, 15)$$

$$f = \Sigma(0, 4, 5, 6, 8, 9) + d(11, 12, 13, 14, 15)$$

for  $a = \sum (0, 2, 3, 5, 6, 7, 8, 9) + d (11, 12, 13, 14, 15)$

0  
2  
3  
5  
6  
7  
8  
9  
10  
11

K map

| CD \ AB | 00 | 01 | 10 | 11 |
|---------|----|----|----|----|
| 00      | 1  | 0  | 1  | 1  |
| 01      | 4  | 1  | 1  | 1  |
| 10      | d  | d  | d  | d  |
| 11      | 1  | 1  | d  | d  |

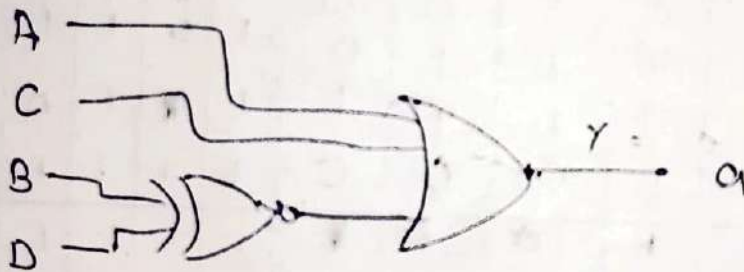
}  $\rightarrow$  octate  
 $\downarrow$   
A

$\downarrow$   
octate.  $\Rightarrow$  C

$\Rightarrow$  20, 20  $\Rightarrow A + C + \bar{B}\bar{D} + BD$

$\downarrow$   
X-NOR gate  
 $\downarrow$

$a = A + C + \overline{B \oplus D}$



for b =  $\Sigma(0, 1, 2, 3, 4, 7, 8, 9) + d(10, 11, 12, 13, 14, 15)$

| CD \ AB | 00 | 01 | 10 | 11 |
|---------|----|----|----|----|
| 00      | 1  | 1  | 1  | 1  |
| 01      | 1  |    | 1  |    |
| 10      | d  | d  | d  | d  |
| 11      | 1  | 1  | d  | d  |

quadrade  $\Rightarrow \bar{A}\bar{B}$

octade  $\Rightarrow A$

quadrade  $\Rightarrow AC$

quadrade  $\bar{C}\bar{D}$

$\bar{C}\bar{D} + \bar{B}C + \bar{B} + \bar{A}B \Rightarrow A + \bar{A}\bar{B} + AC$

for

for c =  $\Sigma(0, 1, 3, 4, 5, 6, 7, 8, 9) + d(11, 12, 13, 14, 15)$

| CD \ AB | 00 | 01 | 10 | 11 |
|---------|----|----|----|----|
| 00      | 1  | 1  | 1  | 1  |
| 01      | 1  | 1  | 1  | 1  |
| 10      | d  | d  | d  | d  |
| 11      | 1  | 1  | d  | d  |

octade  $\Rightarrow$

octade  $\Rightarrow A$

octade  $\Rightarrow \bar{C}$

$$d \Rightarrow 10, 39, 49 \Rightarrow A + \bar{B}\bar{D} + C\bar{D} + \bar{B}C + B\bar{C}D$$

$$e \Rightarrow 29 \Rightarrow \bar{B}\bar{D} + C\bar{D}$$

$$f \Rightarrow 10, 39 \Rightarrow A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

$$g \Rightarrow 10, 39 \Rightarrow A + C\bar{D} + B\bar{C} + \bar{B}C$$

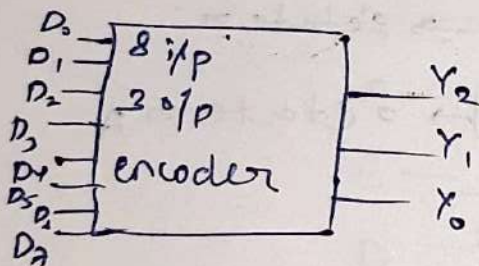
Encoder It is a digital circuit that perform inverse operation of decoder. The opposite of decoding is called encoding.

An encoder is a combinational circuit that convert active input signal into a coded output signal.

### (1) Octal to binary Encoder

input = 8

output = 3



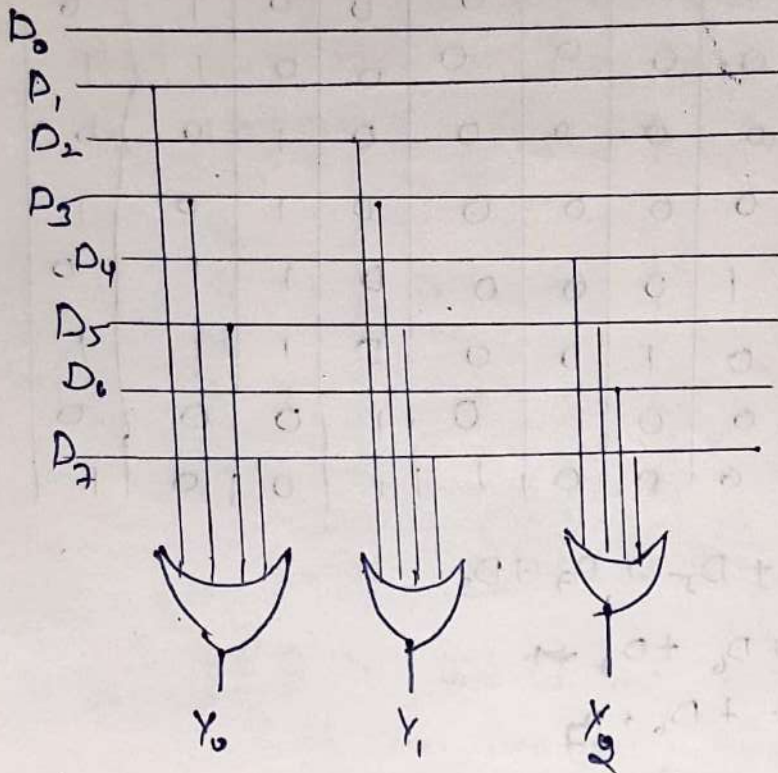
| Input          |                |                |                |                |                |                |                | Output         |                |                |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| D <sub>0</sub> | D <sub>1</sub> | D <sub>2</sub> | D <sub>3</sub> | D <sub>4</sub> | D <sub>5</sub> | D <sub>6</sub> | D <sub>7</sub> | Y <sub>2</sub> | Y <sub>1</sub> | Y <sub>0</sub> |
| 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              |
| 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              |
| 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              |
| 0              | 0              | 0              | 1              | 0              | 0              | 0              | 0              | 0              | 1              | 1              |
| 0              | 0              | 0              | 0              | 1              | 0              | 0              | 0              | 1              | 0              | 0              |
| 0              | 0              | 0              | 0              | 0              | 1              | 0              | 0              | 1              | 0              | 1              |
| 0              | 0              | 0              | 0              | 0              | 0              | 1              | 0              | 1              | 1              | 0              |
| 0              | 0              | 0              | 0              | 0              | 0              | 0              | 1              | 1              | 1              | 1              |

## Circuit diagram

$$Y_0 = D_0 + D_2 + D_4 + D_6$$

$$Y_1 = D_1 + D_3 + D_5 + D_7$$

$$Y_2 = D_4 + D_5 + D_6 + D_7$$

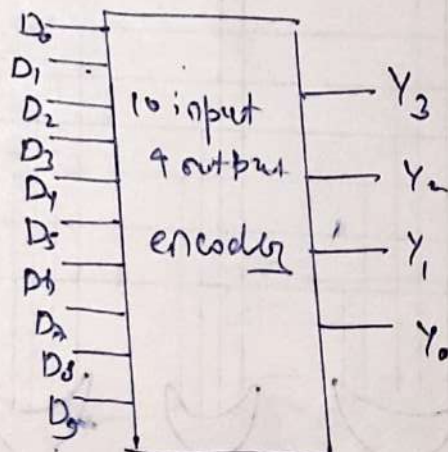


## 122 Decimal to BCD encoder

<sup>\*</sup> (code converter)

input  $\Rightarrow 10$

output  $\Rightarrow 4$



| Input |       |       |       |       |       |       |       |       |       | Output |       |       |       |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--------|-------|-------|-------|
| $D_0$ | $D_1$ | $D_2$ | $D_3$ | $D_4$ | $D_5$ | $D_6$ | $D_7$ | $D_8$ | $D_9$ | $Y_3$  | $Y_2$ | $Y_1$ | $Y_0$ |
| 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 0     |
| 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0      | 0     | 0     | 1     |
| 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0      | 0     | 1     | 0     |
| 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 0      | 0     | 1     | 1     |
| 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0     | 0      | 1     | 0     | 0     |
| 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0     | 0      | 1     | 0     | 1     |
| 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0     | 0      | 1     | 1     | 0     |
| 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0     | 0      | 1     | 1     | 1     |
| 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 0      | 1     | 0     | 0     |
| 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 1     | 1      | 0     | 0     | 1     |

$$Y_0 = D_1 + D_3 + D_5 + D_7 + D_9$$

$$Y_1 = D_2 + D_3 + D_6 + D_7$$

$$Y_2 = D_4 + D_5 + D_6 + D_7$$

$$Y_3 = D_8 + D_9$$

